Where LLM Annotation Substitutes for Human Labelers — and Where It Doesn't: Uncertainty Channels and the Economics of Routing Contested Items

Yogya Mehrotra
yogyamehrotra@gmail.com
github.com/yogyam/DisagreeBench

Abstract

The case for replacing human annotators with LLMs typically rests on a single aggregate agreement number. We test the hypothesis that this number is structurally misleading: that LLM–human agreement is high exactly where humans already agree with each other, and collapses where they don't. Using ChaosNLI (3,113 SNLI/MNLI items with 100 human annotations each), we elicit a label *distribution* from claude-sonnet-5 via self-consistency sampling (10 samples per item) and compare it to the human distribution with Jensen–Shannon divergence (JSD), stratified by human label entropy. The hypothesis holds: mean JSD rises monotonically from 0.07 [95% CI 0.06, 0.09] in the lowest-entropy quintile to 0.31 [0.30, 0.32] in the highest, while the conventional metric — accuracy against the majority label — declines far more gently ($0.94 \rightarrow 0.56$) and therefore flatters the model precisely on the contested items. We then test whether the model's own uncertainty could route hard items to humans, and find it cannot: the model is fully deterministic across its 10 samples on 85% of items — including most items where humans split near 50/50 — and the rank correlation between model self-consistency entropy and human entropy is $\rho = 0.164$ [0.131, 0.196], decisively below any usefully routable signal; enabling reasoning does not change this. The contrast is stark: the human pool's inter-annotator agreement is Krippendorff's $\alpha = 0.37$; the model's agreement with itself is $\alpha = 0.90$. Yet when asked to *estimate* the annotator distribution directly, the same model produces distributions 3× closer to the human ones (JSD 0.07) whose entropy does clear the routing threshold ($\rho = 0.393$). The model's behavioral uncertainty carries no information about human disagreement — but its declarative estimate of human disagreement does. Finally, we run the budget-allocation experiment this signal enables: a simulated router that sends the top-scoring items to k human annotators and keeps the model's distribution for the rest. Against the realistic baseline (model self-consistency distributions), routing on verbalized entropy captures 37% of an expected-gain oracle's achievable improvement, and random allocation needs $\sim 1.5\times$ the annotation budget to match it. Against the strongest baseline (the model's own verbalized distributions), the same signal is *worse than random*: the model's residual errors hide in items it is confidently wrong about. Model-reported uncertainty tells you where the task is hard — not where the model is wrong.

1 Motivation

Data-labeling providers are being asked to justify themselves against claims of the form "the model agrees with human labels 90% of the time at 1% of the cost." That figure is an average over a distribution of examples that is sharply bimodal in difficulty: most items in a real annotation stream are unambiguous, while a minority are genuinely contested — and the contested minority consumes most of a real annotation budget and drives most disagreement in downstream training. If LLM agreement concentrates on the easy majority, a headline agreement figure can be technically true and operationally misleading, and the correct architecture is neither "replace humans" nor "keep humans" but *triage*. This study measures the premise of that argument directly, and then tests whether the triage router is buildable from the model's own uncertainty.

2 Data

ChaosNLI (Nie et al., 2020) provides $\sim$100 annotations per item for 4,645 development-set items from SNLI, MNLI, and Abductive NLI. We use the SNLI (1,514) and MNLI (1,599) portions — 3,113 items sharing the 3-way entailment/neutral/contradiction label space — and exclude Abductive NLI (a 2-way task). The 100-

annotator sample gives a genuine label distribution per item rather than a noisy 3–5-rater estimate, which is what makes entropy stratification statistically meaningful. Items are bucketed into quintiles of human label entropy computed from the full distribution (quintile edges: 0.66, 0.91, 1.05, 1.21 bits).

The pool is genuinely contested: nominal Krippendorff's α is 0.447 (SNLI), 0.283 (MNLI), 0.367 combined. This is by construction — ChaosNLI drew items whose original 5-rater annotations disagreed — and it is exactly the regime the substitution argument is quietly averaged over.

3 Method

Elicitation. claude-sonnet-5 (Anthropic), accessed via the Message Batches API. Each item is sampled 10 times as an independent request. The label options are presented in randomized order per sample (mitigating option-ordering bias); output is constrained to a JSON schema over the three labels, so every sample parses; extended thinking is disabled. The empirical distribution over the 10 sampled labels is the model's label distribution. Of 31,130 requests, 31,127 returned usable labels (3 errors).

Verbalized arm. As a comparison elicitation (one request per item, full 3,113-item set), the model is asked to estimate how 100 careful annotators would distribute over the three labels, output as integer percentages under the same schema constraint.

Router simulation. Each item's 100 annotations are split 50/50: one half defines the ground-truth distribution, the other is the finite pool routed items draw from (so the router can never "buy" the evaluation target, and even perfect routing pays annotator-sampling noise). A router scores all items, sends the top fraction f to humans — receiving k annotations drawn without replacement from the pool — and keeps the model's distribution for the rest. We sweep f from 0 to 1 and $k \in \{1, 3, 5, 10, 20, 50\}$, with 20 replicate splits. Strategies: random; self-consistency entropy; verbalized entropy; channel disagreement (JSD between the sampled and verbalized distributions); an *entropy oracle* scoring by true human entropy; and two regret oracles scoring by each item's actual improvement from routing — *expected gain* (Monte-Carlo mean over 200 redraws; the fair upper bound, since it knows which items benefit but not the luck of the draw) and *realized gain* (a skyline that peeks at the draw). Headline numbers: AUC of the budget–JSD curve, normalized as the share of the expected-gain oracle's improvement over random ("capture"); and budget-to-match, the multiple of annotations random allocation needs to reach a strategy's JSD at a given budget. Item-level bootstrap (1,000 resamples) gives CIs on strategy–random gaps. Everything is repeated with total variation distance in place of JSD and with a bootstrap-resampling split protocol in place of the 50/50 split.

Quintile	n	Mean JSD [CI]	Acc. vs. maj. [CI]
Q1 (low)	631	0.074 [.064, .085]	0.945 [.926, .962]
Q2	615	0.178 [.165, .192]	0.805 [.772, .836]
Q3	623	0.244 [.232, .255]	0.674 [.637, .711]
Q4	622	0.263 [.254, .273]	0.614 [.576, .653]
Q5 (high)	622	0.311 [.302, .321]	0.558 [.519, .596]

Table 1: Model–human divergence and accuracy against the majority label, by quintile of human label entropy (95% bootstrap CIs).

Metrics. Primary: per-item Jensen–Shannon divergence (base 2) between the model and human label distributions, averaged per entropy quintile. Reported alongside, explicitly as the metric under critique: accuracy of the model's modal label against the human majority label. The routing gate: Spearman rank correlation between model self-consistency entropy and human entropy. All means and the correlation carry 95% percentile bootstrap CIs (10,000 resamples). Contamination diagnostic: agreement of the modal label with the *original* single gold label vs. with the ChaosNLI majority, per quintile.

4 Results

4.1 Agreement degrades with human disagreement — and accuracy hides it

JSD rises monotonically by a factor of $\sim$4.2 from Q1 to Q5 (Table 1, Figure 1). This deliberately replicates the stratified diagnostic Nie et al. (2020) ran on fine-tuned encoders, and the shape has changed in a way that matters: where their models' accuracy fell to chance on contested items, the frontier LLM stays well above it, and its Q1 JSD now reaches their estimated human bound ($\sim$0.06). Accuracy against the majority declines much more slowly, and its floor is deceptive: on

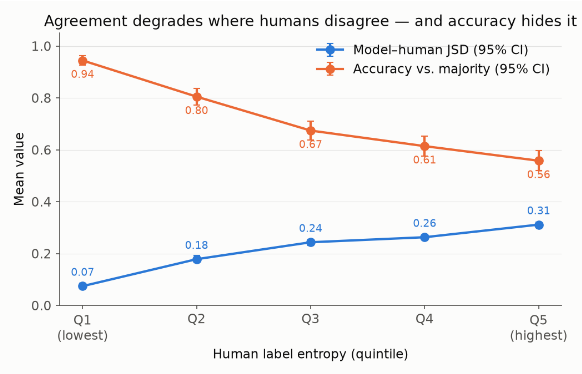

Figure 1: Mean JSD between model and human label distributions (blue) and accuracy against the majority label (orange) across human-entropy quintiles. The divergence quadruples while accuracy declines gently: the standard metric is most flattering exactly where the model is furthest from the human distribution.

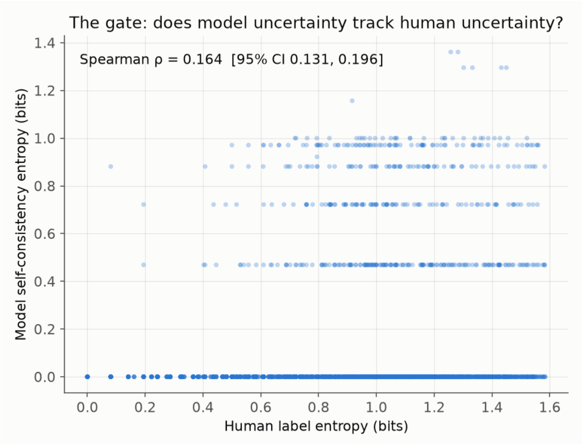

Figure 2: Model self-consistency entropy vs. human label entropy. The model sits at zero entropy on 85% of items regardless of how contested they are; $\rho = 0.164$.

Q5 items the human majority is itself close to a coin flip, so "56% accuracy" largely reflects picking the more common side of a split — not agreement with what humans collectively believe. The gap between the two curves is the headline finding: **the standard metric is most flattering exactly where the model is furthest from the human distribution.**

4.2 The gate fails: model uncertainty is not human uncertainty

If a model knew which items humans find hard, its uncertainty could route those items to people. It does not. Across 10 samples the model returns the identical label on 2,653 / 3,113 items (85%) — including 84% of Q5, where human annotators split at more than 1.2 bits of entropy. Where the human pool agrees at $\alpha = 0.367$, the model agrees with itself at $\alpha = 0.904$.

The rank correlation between model self-consistency entropy and human entropy is $\rho = 0.164$ [95% CI 0.131, 0.196] ($p \approx 4 \times 10^{-20}$; Figure 2). The correlation is reliably nonzero — the model is not blind to difficulty — but the CI excludes by a wide margin any threshold ($\geq \sim 0.3$) at which uncertainty-based routing would meaningfully beat random allocation. An ablation with adaptive thinking enabled (§6) leaves the picture unchanged. By the pre-registered logic of the study design, self-consistency entropy is disqualified as a router input: **this routing signal does not exist** (§4.5 confirms end-to-end that routing on it is no better than random). The signal that does exist comes from a different channel entirely (§4.3).

Signal	JSD	Entropy (bits)	Gate ρ
Sampled (10×)	0.224	0.089	0.164
Verbalized	**0.072**	**0.863**	**0.393**
Human pool	—	0.939	—

Table 2: Mean JSD vs. the human distribution, mean entropy, and Spearman correlation with human entropy, for the two elicitation channels (full set, $n = 3{,}113$; verbalized gate $p \approx 10^{-113}$).

4.3 Verbalized confidence: the model can report disagreement it does not enact

Prompted instead to *estimate* the human distribution directly — "imagine 100 careful annotators; how many choose each label?" (one request per item) — the model behaves entirely differently (Table 2).

Verbalized distributions are $\sim 3\times$ closer to the human distribution than sampled ones, their average entropy nearly matches the human pool's, and verbalized entropy clears the pre-registered routing threshold ($\rho = 0.393 \geq 0.3$; pilot estimate was 0.354). Where sampled JSD degrades $4.2\times$ from the lowest- to highest-entropy quintile (§4.1), verbalized JSD is nearly flat: $0.044 \rightarrow 0.103$. The declarative estimate stays accurate on precisely the contested items where the behavioral distribution collapses. Strikingly, the two uncertainty signals are essentially uncorrelated with each other ($\rho = 0.03$–0.08, depending on which sampled run is paired): they measure different quantities. Together with §4.2, this yields the paper's sharpest formulation: **the model's *behavioral* uncertainty carries no information about human disagree-**

ment, but its *declarative* estimate of human disagreement does. This inverts a common assumption (including in our own study design) that verbalized probabilities are poorly calibrated and self-consistency is the trustworthy signal. It also conditionally revives the triage router: the routing signal exists — it just has to be asked for, not observed.

The direction of this contrast — verbalized beats sampled, and reasoning does not close the gap — independently replicates what Ni et al. (2026) report on hate-speech, emotion, and preference tasks and Meister et al. (2025) on opinion surveys; §5.3 details what is new here (the entropy-flatness, the channel independence, and the routing consequences in §4.5).

One caveat attaches specifically to this result: ChaosNLI's label *distributions* have been public since 2020 — and are by now an explicit optimization target in the literature (e.g., Wu et al., 2026, fine-tunes directly on them) — so the model may have absorbed some of them in pretraining. A memorized distribution would inflate verbalized performance in a way the gold-label diagnostic in §4.4 does not cover. Replication on held-out or post-cutoff annotation data is required before the verbalized result is load-bearing.

4.4 The result is not memorization

ChaosNLI is built on SNLI/MNLI, which are almost certainly in the model's pretraining data; a model could look good by reciting original gold labels. The diagnostic says otherwise: the modal label agrees with the ChaosNLI majority *more* than with the original gold label in Q1–Q3 (e.g. 0.945 vs. 0.857 in Q1), and the two converge in Q4–Q5 (0.614 vs. 0.637; 0.558 vs. 0.539). If the model were reproducing memorized gold labels, the pattern would invert on high-entropy items, where the original gold is essentially arbitrary. It does not (Figure 3).

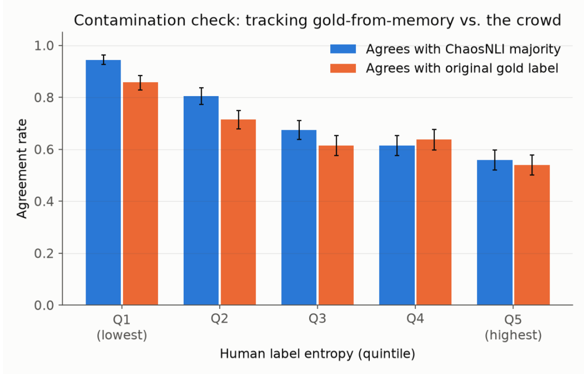

Figure 3: Modal-label agreement with the ChaosNLI 100-annotator majority vs. the original single gold label, by quintile. The model tracks the crowd, not the memorized gold.

4.5 The router: uncertainty finds hard items, not model errors

The simulation (method in §3) asks the operational question directly: given a fixed annotation budget, does routing on model-reported uncertainty beat spending the same budget at random? The answer splits cleanly by what the unrouted items fall back to.

Against the sampled base — the realistic setting where un-routed items keep the model's self-consistency distribution — verbalized-entropy routing works (Figure 4, left). At $k = 10$ annotations per routed item it captures **37.4%** of the expected-gain oracle's improvement over random; channel disagreement (JSD between the model's two distributions, using no human information) captures **45.5%**, statistically indistinguishable from the *true-human-entropy* oracle (47.7%). In budget terms, random allocation needs **1.44–1.57**$\times$ as many annotations to match verbalized-entropy routing at 10–30% coverage (strategy–random gap CIs exclude zero at every budget above 5%). Self-consistency entropy captures nothing (-8.3%), closing the loop on §4.2 with an end-to-end null.

Against the verbalized base — where unrouted items keep the model's verbalized distribution, the stronger baseline §4.3 establishes — the same signal *inverts* (Figure 4, right): verbalized entropy is reliably worse than random (capture -18.7%; random matches it with 0.5–0.6$\times$ the budget). The mechanism is anti-selection. High verbalized entropy flags items where humans genuinely disagree — but §4.3 showed the verbalized distribution is already accurate exactly there. The residual errors sit in low-entropy items the model is confidently wrong about, which entropy routing systematically deprioritizes (per-item verbalized entropy vs. verbalized error: $\rho \approx -0.07$). Even the true-human-entropy oracle barely beats random here (11.9%), while the expected-gain oracle still improves substantially — the errors are findable in principle, just not by any entropy-shaped signal. **Model-reported uncertainty tells you where the task is hard, not where the model is**

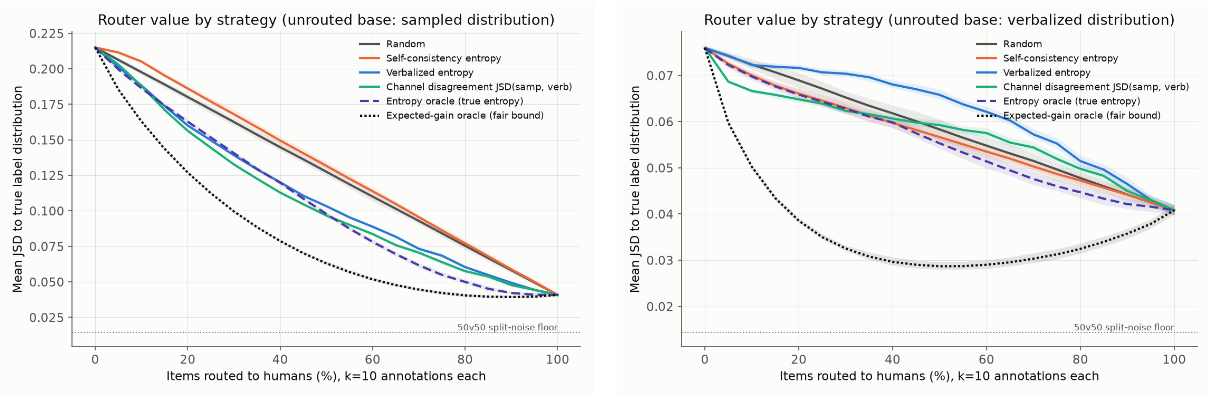

Figure 4: Router value curves at $k = 10$ annotations per routed item. **Left:** unrouted items keep the model's sampled distribution — verbalized-entropy and channel-disagreement routing approach the true-human-entropy oracle. **Right:** unrouted items keep the model's verbalized distribution — the same verbalized-entropy signal is now *worse than random* (anti-selection), and even the true-entropy oracle barely beats random; the expected-gain oracle shows the errors remain findable in principle.

wrong; it routes usefully only when the fallback annotator is weak.

Breadth vs. depth. Sweeping k (Figure 5) shows that at a fixed total budget, shallow-and-broad beats deep-and-narrow on the sampled base ($k = 3$–5 dominates $k = 20$–50), and yields the study's most quotable number on the verbalized base: replacing the model's verbalized distribution with the empirical distribution of k human labels only breaks even at $k \approx 5$ ($k = 1$: JSD 0.270; $k = 3$: 0.113; $k = 5$: 0.072; model verbalized: 0.076). In distributional terms, one model call is worth roughly five human annotations on this dataset — with the memorization caveat of §4.3 attached.

All orderings and magnitudes are stable under total variation distance in place of JSD and under the bootstrap split protocol (capture 37.7–38.9% verbalized, 44.0–46.8% channel disagreement on the sampled base; anti-selection persists on the verbalized base in every variant).

5 Related Work

5.1 Human label variation and distributional evaluation

That disagreement in NLI judgments is reproducible signal rather than annotation noise was established by Pavlick and Kwiatkowski (2019), and the position that label variation should be modeled rather than adjudicated away is now a program (Plank, 2022; Uma et al., 2021; Basile et al., 2021; Cabitza et al., 2023). Our evaluation toolkit is theirs: JSD against the human label distribution and the correlation between per-item model and crowd entropy are both standardized in Uma et al. (2021). Our dataset paper, ChaosNLI (Nie et al., 2020), already performed the stratified analysis for fine-tuned encoders: accuracy degrades toward chance and JSD stays above 0.2 as human agreement falls. §4.1 is a deliberate replication of that diagnostic on a frontier LLM, and what changed is the shape: the model now reaches the estimated human JSD bound ($\sim$0.06) on high-agreement items, and accuracy on contested items no longer collapses to chance (0.56 vs. 0.33) — which makes the flattery problem *worse*, because the standard metric now looks respectable everywhere while distributional divergence quadruples. Distributed NLI (Zhou et al., 2022) formalized the task of predicting human opinion distributions on ChaosNLI: their supervised estimators (MC Dropout, recalibration, distillation) reach JSD $\approx$ 0.18–0.19, which our zero-supervision verbalized elicitation beats by $\sim$2.5$\times$; their appendix also documents that encoder uncertainty is miscalibrated against human disagreement, a precursor of our gate analysis. Baan et al. (2022) showed on ChaosNLI that calibration to a majority label is incoherent under disagreement (our 50/50 truth/pool split in §3 adapts their device of splitting the 100 annotations into subpopulations); Baan et al. (2024) argue conceptually that a predictive distribution conflates model confidence with human variation — our channel-decorrelation finding (§4.3) turns that distinction into a measured property. VariErr NLI (Weber-Genzel et al., 2024) implies some of ChaosNLI's

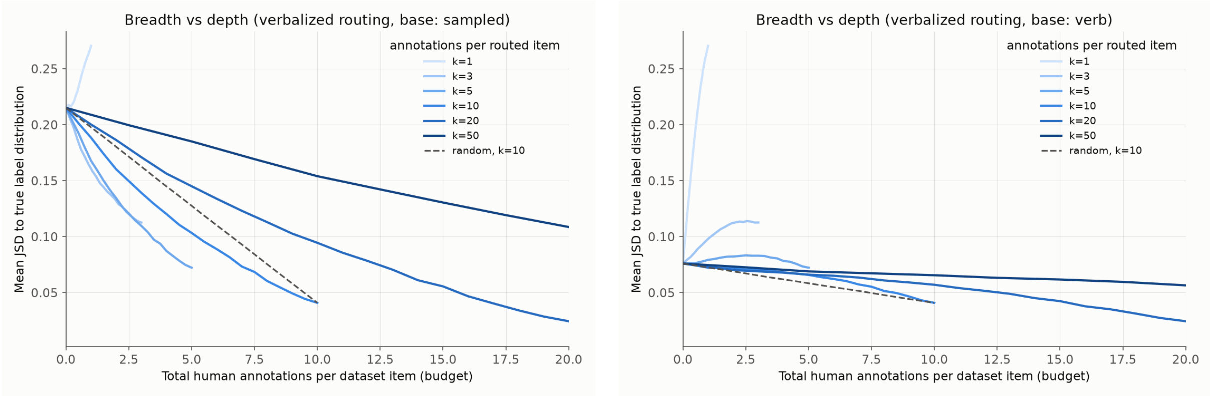

Figure 5: Breadth vs. depth: value of verbalized-entropy routing as a function of total annotation budget, for different annotation depths k per routed item. **Left:** sampled base — shallow-and-broad ($k = 3$–5) dominates deep-and-narrow. **Right:** verbalized base — replacing the model's verbalized distribution with the empirical distribution of k human labels only breaks even at $k \approx 5$; at $k = 1$ routing makes the corpus estimate strictly worse.

variation is annotation error, which makes our router's measured gains conservative with respect to error-cleaned targets.

5.2 LLMs as annotators

The replacement wave (Gilardi et al., 2023; Törnberg, 2023) reads the model's high self-consistency as annotation quality; on contested items we show the same property is a liability — the model agrees with itself ($\alpha = 0.90$) about questions on which 100 humans split nearly evenly ($\alpha = 0.37$). Reiss (2023) complained that 2023-era ChatGPT was too *unstable* to annotate reliably; 2026 frontier models exhibit the opposite pathology, and neither regime tracks human disagreement. Baumann et al. (2025) quantify the downstream stakes — LLM annotation choices flip roughly a third of tested conclusions — and already propose verbalized-confidence-targeted human annotation as a mitigation, evaluated against disagreement-filtered gold labels; our anti-selection result (§4.5) exposes a failure mode of exactly that strategy once the target is the label distribution and the fallback is the model's own best output.

5.3 LLM uncertainty channels and human disagreement

Lee et al. (2023) first compared sampled and logprob-derived LLM distributions to ChaosNLI's annotator distributions, observing that GPT-3's output entropy is near zero and visibly uncorrelated with human entropy. Our §4.2 quantifies that observation at frontier scale (85% exact determinism; $\alpha = 0.90$ vs. 0.37; gate $\rho = 0.164$ with CIs) and stress-tests it against reasoning and paraphrase confounds. Madaan et al. (2025) stratify accuracy by human entropy for open-weight models on ChaosNLI; the accuracy/JSD *dissociation* across quintiles is ours. The channel contrast we find was anticipated on other tasks: Ni et al. (2026) show verbalized distributions beat sampled ones for disagreement prediction on hate-speech, emotion, and preference tasks (and that RLVR-style reasoning hurts); Meister et al. (2025) found LLMs describe opinion distributions better than they simulate them; Jang et al. (2026) attribute the simulation failure to alignment-induced mode collapse (cf. Kirk et al., 2024), consistent with the calibration literature where asking beats token probabilities (Lin et al., 2022; Tian et al., 2023; Kadavath et al., 2022) — though against *correctness*, consistency-based signals often win (Xiong et al., 2024), underlining that our target is different. Wang et al. (2024) show first-token probabilities diverge from what instruction-tuned models actually answer, which is why we omit a logprob arm. On NLI specifically, Chen et al. (2024, 2025) approximate human judgment distributions with explanation-conditioned LLM prompting, and Chen et al. (2026) show CoT does not calibrate distributions on ambiguous items. Relative to all of these, our §4.3–4.4 contribute: replication of the verbalized–sampled gap on 100-way NLI distributions with a frontier model; the near-flatness of verbalized JSD across disagreement levels; the statistical *independence* of the two channels' entropies ($\rho \leq 0.08$), which no prior work reports; and the gate framing that connects channel quality to routability. Zhang et al. (2025) propose verbalized sampling to recover output diversity; our results externally validate the verbal channel against ground-truth human distributions rather than diversity alone.

5.4 Hybrid annotation and budget allocation

Routing between model and human annotators instantiates learning-to-defer (Madras et al., 2018; Mozannar and Sontag, 2020; DeSalvo et al., 2025), but that literature assumes a trained rejector and single-label accuracy; we ask which zero-shot LLM signal could support deferral at all, with distributional fidelity as the objective. CoAnnotating (Li et al., 2023) routes on prompt-paraphrase entropy against single gold labels and reports scalar verbalized confidence unreliable — the opposite ordering from ours, which dissolves once the elicitation target is distinguished: they elicit scalar self-confidence scored against gold labels; we elicit a full annotator distribution scored against real human distributions. HyPAC (Zeng et al., 2026) gives PAC-guaranteed routing for 0-1 error against objective answers; production systems (Kim et al., 2024; Bachar et al., 2026) route escalation on uncertainty and find raw signals wanting. Gligorić et al. (2025) allocate human budget by verbalized confidence for population-level estimates; Mehrotra et al. (2026) do so at demographic-group level; Hakimi et al. (2026) find uncertainty-driven selection fails to beat random in active learning; Schroeder et al. (2025) show humans anchor when *verifying* LLM labels, which motivates our design choice of independent annotation on routed items. Closest to our simulation: Klugmann et al. (2024) route items between crowd-trained soft-label predictors and humans in vision, and Peale et al. (2026) run budget-swept uncertainty-decomposition routing on ChaosNLI with trained classifiers. Kohli (2026) shows on ChaosNLI that annotations needed per item is metric-dependent (distributional metrics saturate near $N = 10$), consistent with our k-sweep; Gruber et al. (2025) pose the annotator-selection question — whether a human or a model provides each label — as an open empirical problem. No prior work runs the experiment in §4.5: an item-level budget sweep over both coverage and annotation depth against real 100-way label distributions, with regret oracles, comparing elicitation channels as routing signals *and* as fallbacks. The fallback-channel reversal — the same uncertainty signal beats random against a weak fallback and anti-selects against a strong one — appears in none of the above.

6 Limitations

- **Elicitation is not the explanation (ablated).** Because thinking was disabled and output schema-constrained in the main run, we replicated the 200-item pilot with adaptive thinking enabled and a 2,048-token budget. Determinism was unchanged (84% vs. 87%), model self-agreement was unchanged ($\alpha = 0.896$ vs. 0.904), and the gate correlation did not improve ($\rho = 0.066$ $[-0.06, 0.19]$, n.s. at $n = 200$). Notably, the model — free to reason at its own discretion — mostly declined to (mean 21 output tokens/sample vs. 11 without thinking): it does not even *perceive* these items as hard. The determinism result therefore survives its most obvious confound, though we cannot rule out that stronger forced-reasoning prompts would behave differently.

- **One model, one task type.** Results are scoped to claude-sonnet-5 on NLI ambiguity; subjectivity-driven disagreement (hate speech, irony — LeWiDi) may behave differently, and is planned as follow-up work.

- **Sampling temperature is not tunable** on this model (the API rejects non-default sampling parameters), so the temperature-sensitivity analysis in the original design is moot; sampling ran at the provider default.

- **10 samples bound measurable entropy.** With 10 draws the minimum nonzero entropy is $\sim$0.47 bits, coarsening the model-entropy scale. This attenuates the gate correlation somewhat; it cannot explain 85% exact determinism.

- **The router is a simulation, not a deployment.** Routed items draw annotations from ChaosNLI's own pool, so annotators are exchangeable with the evaluation target by construction; a real deployment adds annotator drift, and its ground truth would not be a held-out half of 100 labels. The memorization caveat (§4.3) also propagates: if verbalized distributions are partly recalled, both the verbalized base and the routing signal are optimistic. The sampled-base conclusions are less exposed, since the sampled distributions show no memorization signature (§4.4).

- **Prompt sensitivity: conclusions stable, but the sampled signal itself is noisy.** Rerunning the pilot under two prompt paraphrases leaves every headline quantity essentially unchanged (determinism 80–87%; mean JSD 0.21–0.23; gate $\rho = 0.09$–0.12, never approaching 0.3; modal labels agree across prompts on 94% of items). However, *per-item* self-consistency entropy correlates only moderately across paraphrases (Spearman 0.38–0.46) — the little sampling variance that exists is substantially prompt-specific noise rather than a stable property of the item, a further reason it cannot serve as a routing signal.

7 Implications

For the substitution debate, the results assemble into one architecture with a warning label. The stratified curve (§4.1) says the cheap examples are the ones the model already labels well — consistent with aggressive automation of the easy majority. The failed gate (§4.2) says the obvious triage signal — "let the model flag what it's unsure about," read off its sampling behavior — does not exist. The verbalized result (§4.3) supplies the repair: the signal is a *declarative* capability that must be asked for, and the router built on it (§4.5) delivers real budget savings (∼1.5× annotation efficiency) over random allocation in the realistic setting. The warning label is the anti-selection result: the same uncertainty signal is worse than useless for auditing the model's own best output, because model-reported uncertainty locates task difficulty, not model error. Practically: use verbalized distributions as the machine annotation, route by uncertainty only to decide *which items get humans at all*, and do not expect the model to tell you where it is wrong — error-finding needs an independent signal. Regardless, headline agreement numbers should be reported stratified by inter-annotator agreement as a matter of course.

Reproducibility

All code, elicitation scripts, and figures are available at `github.com/yogyam/DisagreeBench`. Pipeline: `prepare.py` → `elicit.py` → `analyze.py`; verbalized arm: `verbalize.py`; router simulation: `router.py` (pure simulation on saved elicitation outputs; no further API calls). Elicitation: claude-sonnet-5, Message Batches API (verbalized full set ran synchronously), 10 samples/item, randomized option order, JSON-schema output, thinking disabled, seed 20260805 for v0–v1 randomization and bootstraps, seed 20260819 for the router simulation. Total API cost: ≈\$20–27 (34,243 requests; ∼12.5M input / ∼0.4M output tokens). ChaosNLI data is not redistributed; `prepare.py` regenerates the processed splits from the original release.

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